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# Development and validation of a detailed kinetic model for RP-3 aviation fuel based on a surrogate formulated by emulating macroscopic properties and microscopic structure

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A R T I C L E I N F O

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A B S T R A C T

This work proposed a kinetic model for RP-3, the most widely used military-civilian aviation fuel in China. Four hydrocarbons within the typical size of major components in RP-3, i.e., n-dodecane, iso-dodecane (2,2,4,6,6-pentamethylheptane), decalin and n-butylbenzene, were included in the component palette to improve the ability of the surrogate to mimic the properties related to the molecular weight. Seven properties, cetane number, molecular weight, H/C ratio, threshold sooting index, lower heating value, the proportion of -$CH_3$ and -$CH_2$ in the total carbons were selected as targets aiming at the comprehensive emulation of the chemical and physical propensities of RP-3. By sequential use of the genetic algorithm and a local search method, a surrogate containing 27.44% n-dodecane, 28.81% iso-dodecane, 26.12% decalin and 17.63% n-butylbenzene by mole was formulated. The autoignition delay times of the surrogate were measured using a heated rapid compression machine at pressures of 10, 15, 20 bar and equivalence ratios of 0.5, 1.0 and 2.0 over low-to-intermediate temperature range. Results show that the surrogate can not only emulate the target properties but also the key non-targeted properties. A kinetic model of 3065 species and 11,898 reactions was then developed based on the proposed surrogate to describe the chemical process during the combustion of RP-3. Simulations show that the model can predict the fundamental combustion datasets in the present work and literature satisfactorily, suggesting the rationality and applicability of the model. Rate of production and evolution histories of OH and $HO_2$ were then conducted using the kinetic model to provide insight into the combustion of RP-3. Analysis suggests that negative temperature coefficient behavior also exists in the first stage ignition.



## 1. Introduction

In the past decades, the number of flights in the world is increasing explosively with the acceleration of globalization, and air transport plays an increasing important role in public transport travel and facilitates the communication. The rapid increase of aircrafts makes air transport one of the main growth points of energy consumption and pollutant emission. The further development of the aviation industry will bring great challenge to energy security and environmental protection in near future if no breakthrough in the energy consumption and emissions of aero-engines is achieved. The performance of the aero-engines, including the efficiency and emission, is mainly determined by the combustion process of jet fuels in the combustor, which is a complex interaction process of turbulence and chemistry in confined space. Therefore, it is of great interest to understand the chemical process of combustion of jet fuels in depth. Actually, as the computational approach becomes prevalent in the optimization and design of combustors, the development of accurate combustion kinetic model for jet fuels has been an issue of interest for the community.

Jet fuels, e.g., Jet-A, JP-8 and RP-3 etc., are complex mixtures of thousands of compounds. For these real fuels, directly developing a kinetic model containing all components to describe the combustion chemistry is out of reach at the present kinetics and computational level due to the compositional complexity. Therefore, surrogate fuels (mixtures of few neat hydrocarbons which shares some key properties with real fuel) are usually used to substitute for real fuels in kinetic modeling and thereby simply the work. Obviously,

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**Table 1**
Summary of proposed surrogate fuels for Jet-A 1, Jet-A and JP-8 in recent years.

| Real fuel | Surrogate | Availability of kinetic model | Ref. |
|---|---|---|---|
| Jet A-1 | n-decane/iso-octane/n-propylbenzene (53.2 /21.6/25.2 V.%) | Yes | [14] |
| Jet A/JP8 | n-dodecane/ methylcyclohexane / m-xylene (30.3/48.5%21.2 mole%) | Yes | [7] |
| Jet A POST-4658 | n-dodecane/iso-cetane/ trans decalin/toluene (30/36/24.6/9.4 mole%) | Yes | [15] |
| Jet A POSF-4658 | n-dodecane/iso-cetane/methylcyclohexane/toluene (38.44/14.84/23.36/23.36 mole%) | NO | [16] |
| Jet A POSF-4658 | n-dodecane/iso-cetane /decalin/toluene (28.97/14.24/ 31.88/24.91 mole%) | NO | [16] |
| Jet A POSF-4658 | n-dodecane/2,5–2methylhexane/toluene (50.9/21.9/27.2% mole%) | Yes | [13] |
| Jet A POSF-4658 | n-dodecane/n-decane/iso-cetane/iso-octane/decalin/toluene (47.06/0/16.69/0/ 24.19/12.06 mole%) | Yes | [8] |
| Jet A POSF-5642 | n-dodecane/n-decane/iso-cetane/iso-octane/decalin/toluene (14.16/0/31.41/ 40.16/14.27/0 mole%) | Yes | [8] |
| Jet A POSF-4658 | n-dodecane/iso-octane/toluene/iso-cetane/decalin (37.06/1.95/25.91/20.59/14.49 mole %) | Yes | [17] |
| Jet A POSF-4658 | decalin/n-dodecane/iso-cetane/iso-octane//toluene (0.5/40.11/12.49/9.8/37.1 mole%) | Yes | [18] |
| Jet A-1 | n-decane/n-propylbenzene /n-propylcyclohexane (76.7/13.2/10.1 wt.%) | Yes | [19] |
| Jet A | n-decane/iso-cetane/ m-xylene (61.11/ 11.93/26.96 mole%) | Yes | [20] |

surrogate formulation is the prerequisite for detailed combustion modeling of jet fuels in this strategy.

Great efforts have been made to the surrogate formulation for jet fuels. In early researches, single-component surrogates (e.g., n-decane and n-dodecane [1–5]) were usually used to approximate the combustion behaviors of jet fuels due to the unavailability of chemical mechanisms for most neat components and the lack of computational resources. However, it is almost impossible to accurately replicate the combustion characteristics of complex real fuels by only a single component. With the development of kinetics and computational level, multi-component surrogate was proposed to emulate real jet fuels. The optimization method based on matching target properties was the mainstream method for multi-component surrogate formulation in the last decade. A representative method is the combustion target property target matching (CPT) method proposed by Dooley et al. [6]. Dooley et al. chose derived cetane number (DCN), molecular weight (MW), hydrocarbon/carbon (H/C) molar ratio and threshold sooting index (TSI) as property targets, and proposed a four-component surrogate for Jet-A POSF 4658. It was found that the proposed surrogate could reproduce the combustion characteristics of the target fuel at a satisfactory level under pre-vaporization conditions. On this basis, Narayanaswamy et al. [7] introduced an optimization process to determine the final surrogate composition and proposed an improved method. In addition, Violi's group [8–10] is always committed to develop surrogates that emulate physical and chemical properties simultaneously. The selected key properties included low heating value (LHV), DCN, H/C, MW, distillation curve, viscosity, specific heat and density and so on. Distinct from the above-mention researches which chose macroscopic properties as matching targets, Jameel et al. [11] proposed the so called minimalist functional group approach which selected micro functional groups as matching target. Based on the concept that both physical and chemical properties are the consequence of molecular structure, they select typical functional group such as -$CH_3$ and -$CH_2$ etc. as target properties. They found that the proposed surrogates were able to replicate the combustion properties of the target fuels. It must be pointed out that this method is very similar to the approach used in the work of Yu and Gou [12,13].

Based on these formulation methodologies, a wide range of surrogates were proposed for jet fuels, and kinetic mechanism were developed for parts of the surrogates. Table 1 illustrated the surrogates proposed and for Jet A, Jet A-1 and JP 8 in recent years, which contributed to the understanding of the combustion process of these real fuels.

As the most widely used military-civilian aviation fuel in China, RP-3 has attracted increasing interest recently. In 2015, Zhang et al. [21] proposed a binary surrogate (n-decane/1,2,4-trimethylbenzene 80/20 wt.%) for RP-3, and the kinetic mechanism for the surrogate was provided as well. It is found that this model captures the high-temperature ignition delay times (IDTs) of RP-3 satisfactory. Yu and Gou [12] developed a ternary surrogate (n-Dodecane/2,5-dime thylhexane/Toluene 54.3/32.1/13.6 mole%) and proposed a kinetic scheme for it. However, in a very recent research [22], these two kinetic models failed to accurately predict the autoignition characteristics over a wider condition range, especially at low temperatures. Wu et al. [23] proposed two surrogates for RP-3 and presented a series of comparison between the surrogates and R-3. Although these surrogates reproduced the propensities of the target fuel well, their kinetic mechanisms were unavailable owing to the inclusion of several components without kinetic mechanism, and thereby limited their application. Very recently, a five-component surrogate containing n-decane/n-dodecane/iso-cetane/methylcyclohexane/toluene (14%/10%/30%/36%/10% by mole) was developed by Liu et al. [24] to emulate both the physical and chemical properties of RP-3 simultaneously. Results showed that this surrogate could mimic the properties of interest at an acceptable level. Furthermore, Li et al. [25] proposed two five-components surrogates composed of n-dodecane, 2,2,4,4,6,8,8-heptamethylnonane, 2-methylheptane, decalin and o-xylene; Liu et al. [26] developed a kinetic model for a tri-component surrogate (n-dodecane/n-propylbenzene/1,3,5-trimethylcyclohexane, 66.2%/15.8%/ /18.0% by mole); Bai et al. [27] suggest two surrogates for RP-3, but the kinetic model showed obvious discrepancy compared with the measured IDTs and speciation profiles. There are also some surrogate fuels proposed to investigate the sooting tendency [28,29].

The above-mentioned kinetic models provided insights into the combustion chemistry of jet fuels, however, there are still some deficiencies in these researches. Due to the complexity of kinetic scheme and the great challenge confront in fundamental combustion experiments, many heavy hydrocarbons have no mature of kinetic model. Therefore, surrogates have no available kinetic mechanism or have to adopted components which are out of the typical molecular size range of major components in jet fuel. For the latter, basically it inevitably would hamper the surrogates' ability to reproduce the propensities related to molecular size. Such phenomena are very common for both kerosene and diesel surrogate. Typically, a proposed surrogate is generally expected to be able to mimic more properties of the target fuel as possible so as to extend its application, except those surrogates only designed for specific application, e.g., only for investigating the polycyclic aromatic hydrocarbons formation [28–31]. According to Kim and Violi [32], appropriate selection of surrogate components benefits the successful surrogate formulation. Therefore, selecting components with similar molecular size to real jet fuel would improve the performance of surrogates.

This study aims to propose a kinetic model of high accuracy for RP-3, including a surrogate and the corresponding chemical reaction mechanism, and provide fundamental combustion experimen-

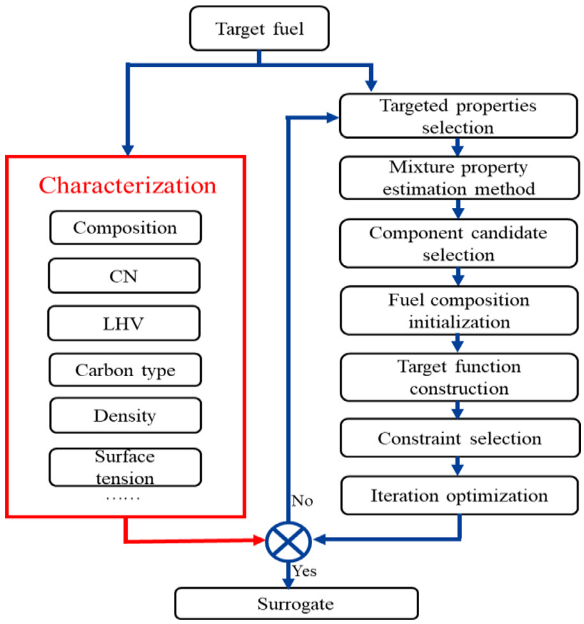


**Fig. 1.** Schematic of surrogate formulation.

tal data to evaluate the model. Both macroscopic properties and microscopic structure are selected as targets to improve the ability of the surrogate to mimic the target fuel. The next generation of surrogate components for jet fuels [33], which fall within the typical molecular weight range of the real fuel components, will be selected as candidate surrogate components to improve the agreement with target fuel. A surrogate would be formulated for RP-3 by employing a regression optimization method, and its kinetic model would be developed in this study as well. IDTs will be measured in a heated rapid compression machine (RCM) at Shanghai Jiao Tong university (SJTU) to evaluate the proposed model.

## 2. Development of the kinetic model

### 2.1. Surrogate formulation

The flow chart of surrogate formulation is illustrated in Fig. 1, which can be divided into four steps: the characterization of the target fuel, target propensity selection, component candidate selection and optimization of the composition. Since the characterization of RP-3 has been conducted in our previous studies [22,23], this section only focuses on the other three steps.

#### 2.1.1. Target propensities

In order to reflect the major properties of RP-3 of interest, it is essential to select the target properties according to the end application of the surrogate. Since the properties related to combustion is multifaceted, the inclusion of only limited macroscopic properties as target properties to be matched is almost impossible to comprehensively match all properties, especially the nontarget properties, but they may exert influence on combustion process as well. The case including only microscopic properties is similar, e.g., the molecular weight is totally unconstrained. Won et al. has demonstrated that the global combustion characteristics of hydrocarbons are closely correlated with the $CH_3/CH_2$ ratio [34]. Hence, the proportions of $CH_3$ and $CH_2$ group are chosen to represent the microscopic molecular structure. Finally, six properties, including CN, H/C ratio, MW, LHV, TSI and carbon types ($CH_3$ and $CH_2$), were selected as target properties to cover both macro and micro properties in this study. Such selection could ensure the consistency of the key macro properties of interest and potentially ensure the similarity of other nontarget properties as far as possible.

#### 2.1.2. Selection of surrogate component candidates

The selection of candidate components determines the potential of the formulated surrogate to reproduce the properties to a great extent. Hence, it is essential to include appropriate components in the surrogate palette. Jet fuels contains a large number of heavy hydrocarbons, and thereby surrogates in literature which were comprised of small molecule hydrocarbons cannot mimic the propensity of jet fuels comprehensively. In this study, components with similar molecular size to practical RP-3, including n-dodecane, iso-dodecane (2,2,4,6,6-Pentamethylheptane), decalin and n-butylbenzene, were selected as candidate components to emulate the properties of RP-3 more comprehensively. The major physical-chemical properties of the candidate components are listed in Table 2.

Parts of these candidates were seldom included in the palette for jet fuels in previous studies due to the immaturity of reaction mechanisms. However, validated kinetic models for these compounds are now available [41–44] with the development of reaction kinetics, which contributes to the development of kinetic mechanism for the proposed surrogate.

#### 2.1.3. Optimization of surrogate composition

A multi-property regression algorithm is used to optimize the surrogate composition. The objective function of the optimization method was defined as the sum of the squared relative difference of target properties between RP-3 and the surrogate, as expressed in Eq. (1):

$$F = \sum_{i=1}^{6} \left( \frac{p_{ic} - p_{it}}{p_{it}} \right)^2 \tag{1}$$

Where, $P_{ic}$ and $P_{it}$ are property $i$ of surrogate fuel and RP-3, respectively, i is the index of target properties.

The target properties of the surrogate are estimated according to the following equations:

$$\mathrm{LHV} = \sum_{i=1}^{N} LHV_i * Y_i \tag{2}$$

$$\mathrm{CN} = \sum_{i=1}^{N} CN_i * V_i \tag{3}$$

$$\mathrm{MW} = \sum_{i=1}^{N} MW_i * x_i \tag{4}$$

$$\frac{\mathrm{H}}{\mathrm{C}} = \sum_{i=1}^{N} (H_i * x_i) / \sum_{i=1}^{N} (C_i * x_i) \tag{5}$$

$$\mathrm{TSI} = \sum_{i=1}^{N} TSI_i * x_i \tag{6}$$

$$CH_3 = \frac{\sum_{i=1}^{N} CH_{3,i} * x_i}{\sum_{i=1}^{N} NC_i * x_i} \tag{7}$$

$$CH_2 = \frac{\sum_{i=1}^{N} CH_{2,i} * x_i}{\sum_{i=1}^{N} NC_i * x_i} \tag{8}$$

Here, $i$ = 1–4, $x_i$, $Y_i$ and $V_i$ are the mole faction, mass fraction and volume fraction of component i, respectively; $LHV_i$, CNi, $MW_i$, $C_i$, $H_i$ and $TSI_i$ are the LHV, CN, MW, carbon number, hydrogen number and TSI of component $i$, respectively; $CH_{3,i}$ and $CH_{2,i}$ is the number of carbon type $-CH_3$ and $-CH_2$ in component $i$, and

**Table 2**
Physical and chemical properties of candidate components.

| Component | CN | LHV (MJ/kg) | MW | H/C | TSI | Surface tension at 25 °C |
|---|---|---|---|---|---|---|
| n-dodecane | 82.5 [24] | 44.11 [9] | 170.4 | 2.17 | 7 [35] | 24.94 [36] |
| Iso-dodecane | 16.8 [37] | 44.14 [23] | 170.4 | 2.17 | 15.4 [37] | 21.58 [23] |
| Decalin | 48 [38] | 42.58 [9] | 138.3 | 1.80 | 22 [35] | 32.17 [23] |
| n-butylbenzene | 14.3 [39] | 41.457 [36] | 134.2 | 1.40 | 62 [40] | 28.63 [36] |

N is the number of candidate components in the surrogate. With the sequential use of a global search method (the genetic algorithm) and a local search method (MATLAB function 'fmincon') to minimize the value of Eq. (1), a four-component surrogate composed of 27.44% n-dodecane, 28.81% iso-dodecane, 26.12% decalin and 17.63% n-butylbenzene by mole was formulated for RP-3.

### *2.2. Development of surrogate kinetic model*

The kinetic reaction mechanism of the surrogate included the C0-C4 core mechanism and the sub-mechanisms of the four components. Due to the kinetic model of components used AramcoMech 2.0 [45] as core mechanism, therefore, for consistency AramcoMech 2.0 was still used as core mechanism. The mechanisms of n-dodecane and iso-dodecane were developed and validated in our recent studies[41, 42], and thereby the sub-mechanisms of the two hydrocarbons were constructed base on these mechanisms. The sub-mechanism of decalin used in this study was based on the model developed by Wang et al. [43], while that of n-butylbenzene was based on the model developed by Nakamura et al. [44]. In order to improve the performance of the n-butylbenzene sub-mechanism in low-to-intermediate temperature range, the activation energy of the keto-hydroperoxide decomposition was increased from 39,000 cal/mole to 42,500 cal/mole, which falls within the range of this reaction class in literature (39,000–44,000 cal/mole) [46,47]. Species in common and duplicate reactions among sub-mechanisms were identified and the resulting incompatibility were resolved appropriately. For the conflict of thermodynamic data of common species, we retained the data from newer sources. While for duplicate reactions with different kinetic data sets, we retained those using rate constants from theoretic calculation or experimental measurement or newer source. Finally, a kinetic model of 3065 species and 11,898 reactions was developed to describe the combustion chemistry of the surrogate.

## 3. Experimental and numerical methods

In this study, the IDT of the surrogate were measured using the heated RCM at SJTU. The experimental procedure had been introduced in detail in our previous studies [22,48], and thus it won't be repeated here. In order to keep the experimental conditions completely consistent with experiments for RP-3 [22], a preheating temperature of 140 °C was used to vaporize the components. Through monitoring the partial pressure, it was found that the vaporization rate of the surrogate was above 99% at all conditions used in this study. In order to simulate the facility effect, a non-reactive run, in which the oxygen was replaced using identical nitrogen, was performed for each condition. The IDT was defined as the time between the end of compression (EOC) and the moment with the maximum pressure derivative, as shown in Fig. 2. According to the variation of the pressure derivative, the total IDT ($\tau$) can be divided into the first-stage IDT ($\tau 1$) and the second-stage IDT ($\tau_2$). The repeatability of the measurements was also presented in Fig. 2. The test conditions used are provided in Table 3. Using the method in literature [49], the systematic uncertainty was estimated to be within ±15%.

Simulations was performed with the Closed Homogeneous Batch Reactor model in Chemkin software to evaluate the kinetic model. Constant volume (CONV) and variable volume (VPRO) simulation were used to calculate the IDTs, respectively. Herein, the variable volume was derived from the pressure traces of non-reactive to mimic the facility effect, including heat loss and compression stroke.

## 4. Results and discussion

### *4.1. Surrogate assessment*

Since the performance of a kinetic model for complex real fuel depends on both the surrogate and the reaction mechanism, it is of great necessity to evaluate the ability of the surrogate before using the surrogate for simulation.

#### *4.1.1. Comparison of key properties of RP-3 and surrogate fuel*

Table 4 shows a comparison of those target properties between RP-3 and the surrogate. On the whole, the surrogate satisfactorily matches the target fuel and almost all properties are very close to RP-3, especially CN and H/C ratio, which are closely related to the combustion characteristics. It is found that the groups -$CH_3$ and -$CH_2$ account for the majority of the carbon types of RP-3, and the two fuels share very similar $CH_3/CH_2$ ratio, i.e., 26.4/38.7 and 24.64/34.6, respectively.

The similarity of these target propensities, especially the molecular structure, promised the ability of the surrogate to replicate those non-target properties. Figure 3 compares some key but non-target propensities of the two fuels, including the density, surface tension and viscosity. These physical properties affect the delivery of fuel in the aeroengine, as well as the vaporization. Specifically, the density is associate with the bulk modulus, and thereby affects fuel injection time, while the surface tension and viscosity are always utilized in the modeling of spray breakup since they determine the Reynolds and Weber numbers. The results show that the two fuels have very close density, viscosity and surface tension under the investigated conditions, indicating that the surrogate has the potential to replicate the performance of RP-3 in practical engines.

#### *4.1.2. Comparison of autoignition characteristics between RP-3 and surrogate fuel*

The IDT of the fuel was greatly related to the combustion efficiency, power output, emission and combustion stability of aero-engine. Hence, a reasonable surrogate must be able to replicate the autoignition characteristics of the target real fuel. Figure 4 presents a comparison of the IDTs between the proposed surrogate and RP-3 to show the ability of the surrogate in reproducing the auto-ignition characteristics of the target fuel in a wide range of operating conditions. It can be observed that the surrogate successfully captured the auto-ignition characteristics of RP-3 in the whole temperature range. Obviously, the IDTs of the two fuels share the same variation tendency, including the temperature window of negative temperature coefficient (NTC) behavior and the variation with pressure and temperature. Besides the trend,

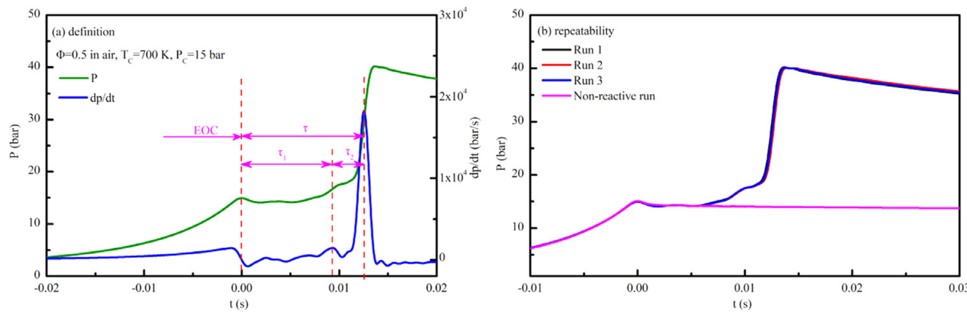


**Fig. 2.** Definition of IDT in RCM measurements and experimental repeatability.

**Table 3**
Test condition used in this study.

| Φ | $x_{Fuel}$ (mole%) | $x_{O2}$ (mole%) | $x_{N2}$ (mole%) | Dilution ratio ($x_{N2}/(x_{N2}+x_{O2})$) | Pressure (bar) |
|---|---|---|---|---|---|
| 0.5 | 0.63 | 20.87 | 78.50 | 79.0% | 10/15 |
| 1.0 | 1.26 | 20.74 | 78.00 | 79.0% | 10 |
| 1.5 | 1.87 | 20.61 | 77.52 | 79.0% | 10 |
| 1.0 | 0.63 | 10.43 | 88.93 | 89.5% | 10/15/20 |
| 1.5 | 0.63 | 6.96 | 92.41 | 93.0% | 10/15/20 |

**Table 4**
comparison of the target properties between RP-3 and the surrogate.

| Fuel | CN | LHV (MJ/kg) | MW | H/C | TSI | $CH_3$ | $CH_2$ |
|---|---|---|---|---|---|---|---|
| RP-3 | 43.30[a] | 42.40[a] | 165.0 | 1.96 | 24.00[b] | 26.40%[a] | 38.70%[a] |
| Surrogate | 43.50 | 43.36 | 155.6 | 1.96 | 23.03 | 24.64% | 34.60% |

a data from Wu et al. [23]; b data from Narayanaswamy et al. [7].

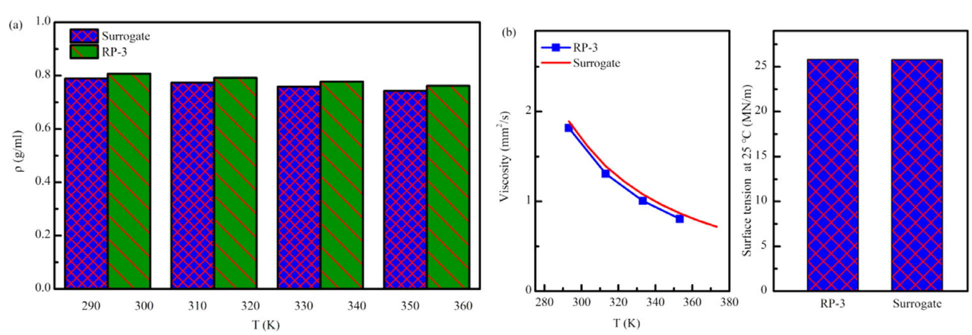


**Fig. 3.** Comparisons of some key non-targeted physical properties for RP-3 [23] and the surrogate.

the magnitude of measured IDTs is almost the same and the divergence basically falls within ±15%.

The IDTs were derived from the raw pressure traces and some useful information may be lost in this process. Therefore, it is essential to compare the raw pressure traces to verify the surrogate. Figure 5 displays a comparison of the raw pressure traces of the two fuels under typical operating conditions. The results show that the surrogate reproduce the pressure histories of RP-3 satisfactorily at all the shown conditions. At these test conditions, the two fuels both show double-stage ignition events. The surrogate captures the first-stage IDT and the first-stage pressure rise of the real fuel perfectly. Furthermore, the peak pressures after hot ignition of the two fuels are very close to each other, as well as the heat loss, which can be observed from the slope of pressure drop. Except slightly longer than RP-3 under test condition shown in Fig. 5(c), the total IDTs of the surrogate are almost identical to those of this real fuel. Take the systematic uncertainty into consideration, it can be concluded that the pressure histories of the two fuels agree very well with each other. The similarity of the pressure histories further demonstrates the appropriateness of the surrogate.

In summary, the autoignition characteristics of the surrogate are very similar to the target fuel, which enable the ability of the surrogate to work as substitute.

### *4.2. Assessment of the kinetic model*

#### *4.2.1. Validation against the neat components*

Typically, in order to ensure that the agreement between the simulation and experiments is not just a coincidence for the surrogate with a specific composition, the combined model must remain the ability to predict the combustion characteristics of neat components. Therefore, the model was used to simulate the neat components to evaluate the fidelity of the combined model to pure components firstly. Figures 6–9 display the comparisons of the predictions from the combined model and the experimental data under typical conditions in literature [37,41–44]. Generally, the model satisfactorily predicted the experimental data for these components, demonstrating that the merging of the sub-mechanisms has little influence on good fidelity of the sub-mechanisms.

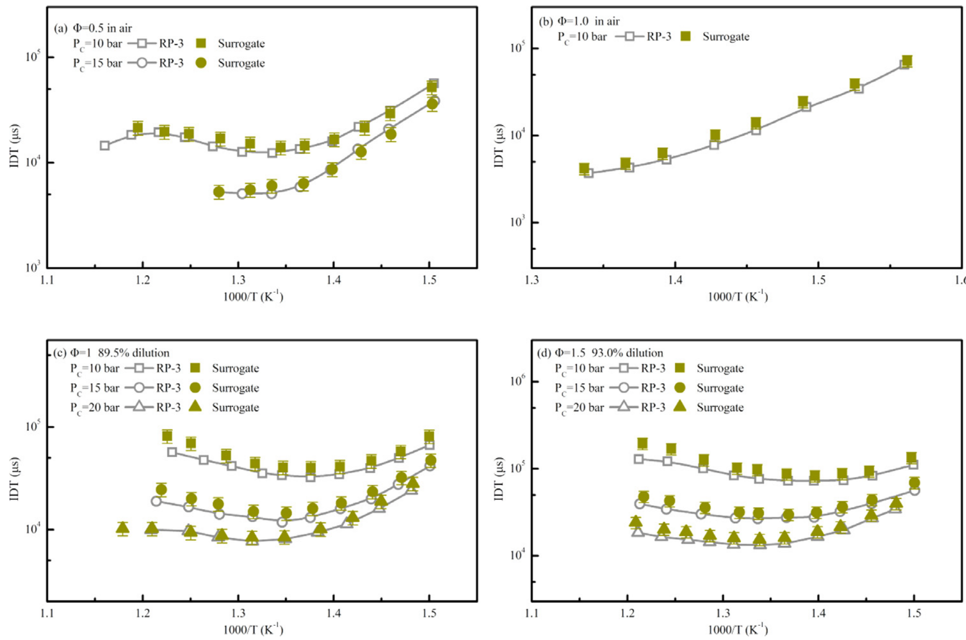


**Fig. 4.** Comprarison of IDTs between RP-3 [22] and surrogate under the same conditions.

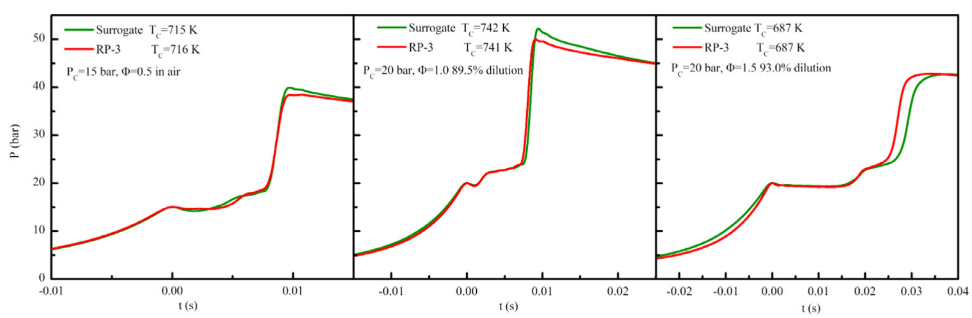


**Fig. 5.** Comprarison of pressure trace between RP-3 [22] and surrogate under the similar conditions.

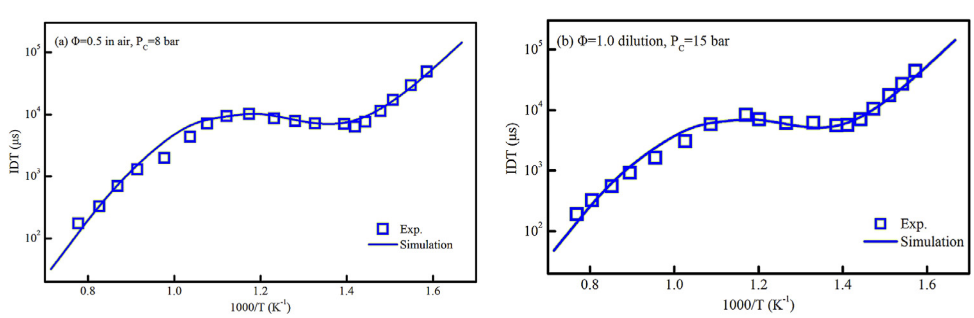


**Fig. 6.** Comparisons of the simulation using the combined model with experimental data of n-dodecane in [42].

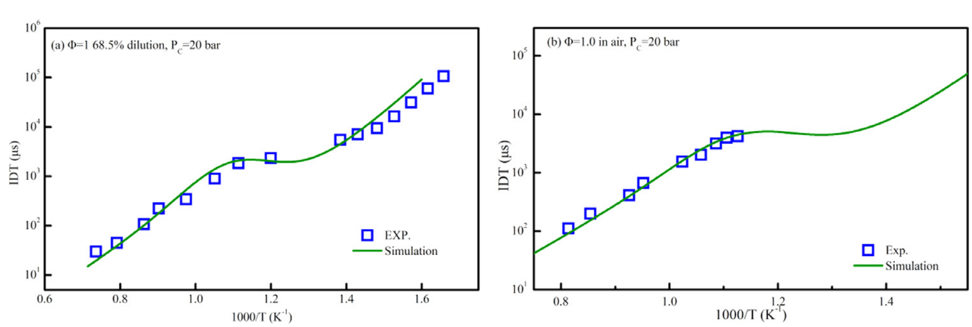


**Fig. 7.** Comparisons of the simulation using the combined model with experimental data of iso-dodecane in [37,41].

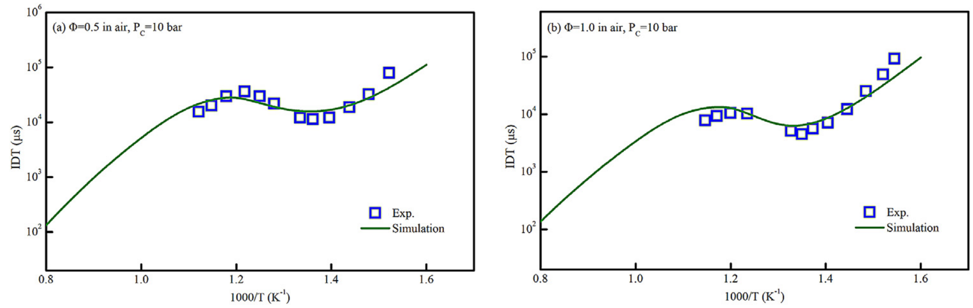


**Fig. 8.** Comparisons of the simulation using the combined model with experimental data of decalin in [43].

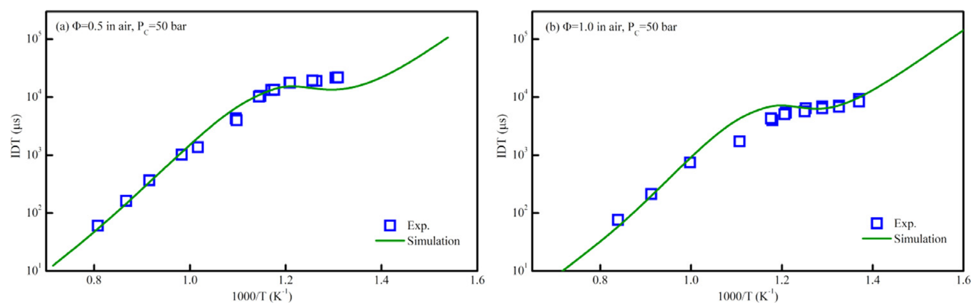


**Fig. 9.** Comparisons of the simulation using the combined model with experimental data of n-butylbenzene in [44].

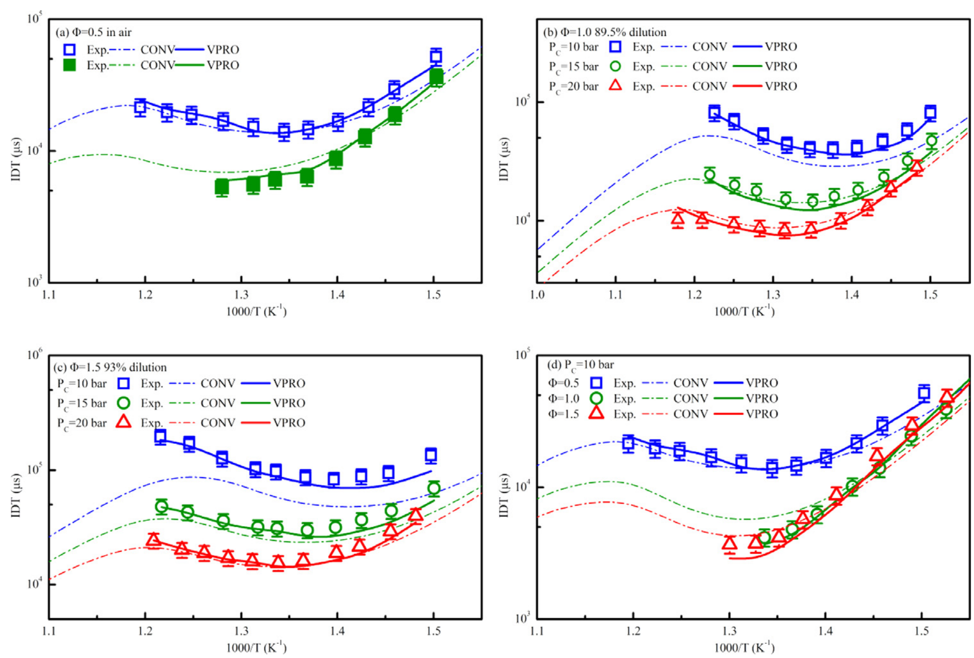


**Fig. 10.** Comparisons of measured and predicted IDTs of the surrogate under different equivalence ratio.

*4.2.2. Evaluation of the kinetic model based on the autoignition characteristics of the surrogate*

In this section, the kinetic model was used to simulate the IDTs of the surrogate using Chemkin software. Both CONV and VPRO simulation were conducted to evaluate the reaction mechanism. Figure 10 compares the experimental and calculated IDTs of the surrogate over wide pressure and equivalence ration ranges. Since the aeroengine need to work reliably over a very wide dilution ratio range when the flight altitude varied from the ground to more than 10,000 m, different dilution ratios were used to evaluate the performance of the model under diluted especially extremely diluted conditions. Panel (a)–(c) in Fig. 10 shows the dependent of IDTs on the pressure for different reactant mixtures. As expected, the reactivity of mixtures increases with increasing pressure, and the effect of pressure become most pronounced in the NTC region because the pressure-dependent reaction $H_2O_2$+(M)=2OH+(M) plays an increasing important role in this temperature range. The model accurately captures this trend. Figure 10(d) presents the effect of equivalence ratio on IDTs. With the increase of fuel, the IDTs become shorter significantly on the fuel lean side; while on the fuel rich side, the IDTs only show very weak sensitivity to the fuel concentration. Similar phenomena was observed in some pure hydrocarbons [50].

Furthermore, the model well replicates the temperature window of NTC behavior under all testing conditions. The variation of predicted IDTs with temperature are almost fully synchronized

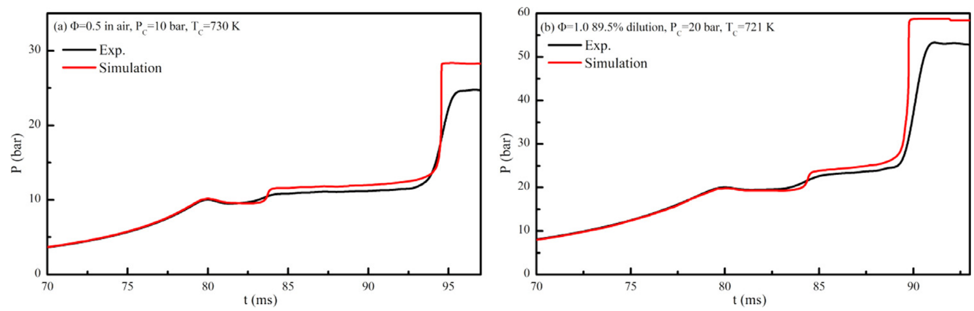


**Fig. 11.** Simulated and measured pressure trace under typical conditions.

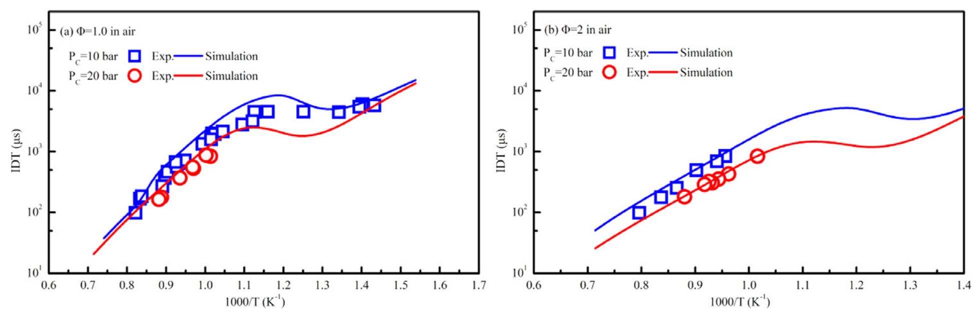


**Fig. 12.** The experimental and simulated IDTs for RP-3 [21].

with the measured ones. On the whole, the kinetic model not only well follows the trend of IDTs with pressure and equivalence ratio, but also the magnitude of the experimental data. Generally, the predicted IDTs using VPRO simulations which take the heat loss and compression stroke into account agree well with the experimental data, and the discrepancies basically fall within the error bar of 15%.

The predicted pressure traces using the volume history were further used to compare with the measured ones, as shown in Fig. 11. It is found that the model accurately predicts the first stage and total IDTs. Before the pressure rise of the first stage ignition, the two profiles almost overlap with each other completely. Since the simulation was performed using the volume histories of the non-reactive runs whose temperatures were lower than the reactive ones after the first stage ignition, the simulated heat loss tends to be smaller than the actual one. Therefore, the simulated pressures are slightly higher than the experimental ones after the first stage ignition. Similar situations are observed for the pressure rise rate.

On the whole, the kinetic model provides satisfactory predictions for the autoignition characteristics of the surrogate. The good agreement demonstrates that the kinetic model can be used to describe the autoignition behavior of RP-3, given the almost same autoignition characteristics between the surrogate and RP-3 as shown in the previous section.

*4.2.3. Validation of the kinetic model against data in literature*

To assess the performance of the kinetic model more comprehensively, fundamental combustion experimental data for RP-3 in literature was used as target to evaluate the model. Zhang et al. [21] measured the IDTs of RP-3/air mixtures using a heated shock tube. Simulations were conducted for this dataset using the closed homogenous batch reactor model with constant volume constraint. According to the suggestion of the authors, a pressure rise of 3%/ms was taken into account. As shown in Fig. 12, the model well follows the variation trends of the IDTs with temperature and pressure under the test conditions, except for slightly overpredicting the IDTs at intermediate temperatures for equivalence ratio of 1.0 in air at 10 bar. Considering the experimental uncertainty, this discrepancy is acceptable.

Li et al. [25] investigated the speciation of oxidation of RP-3 in a plug flow reactor using AR as the balance gas. This dataset was also used to assess the proposed model, as shown in Fig. 13. Overall, the model well captured the trend of the major species with temperature during the oxidation. For equivalence ratio of 0.6, a weak NTC behavior was found in terms of the consumption of $O_2$ and the formation of $C_3H_6$. The model predicted this behavior; however, it must be pointed out that the calculated NTC region shifts to a lower temperature range than the experimental results. Typically, the temperature distribution along the axis is not uniform but trapezoidal distribution. Therefore, the actual temperature profiles is needed to mimic this inhomogeneity. However, the temperature profiles were not provided in [25], and the boundary condition have to be set to uniform in the simulation. Hence, such a discrepancy can be expected.

## 5. Kinetic analyses

NTC behavior is a significant characteristic for heavy hydrocarbons and is closely related to the combustion performance of engines. Since HO2 and OH radicals are believed to plays an important role in NTC region, attentions are paid to the rate of production of these species in this section based on the above proposed kinetic model to give insight into the NTC phenomenon. Rate of production (ROP) analysis of OH and $HO_2$ was performed for fuel/air mixture with equivalence ratio of 1 at temperature of 800 K (a typical temperature in NTC region) under pressure of 10 bar, as shown in Fig. 14. The time of 0.2 ms prior to the first stage ignition was selected for analysis because the reactions were not affected by thermal runaway at this time.

Generally, it is found that the main consumption pathway is the H abstraction reaction from the parent fuel molecules by OH, as shown in Fig. 14(a). The H abstraction reaction from secondary carbon sites of decalin consumes the most OH radicals due to relative weak C–H bond and the abundant identical H atoms at this site attributing to the symmetry of molecules. On the whole,

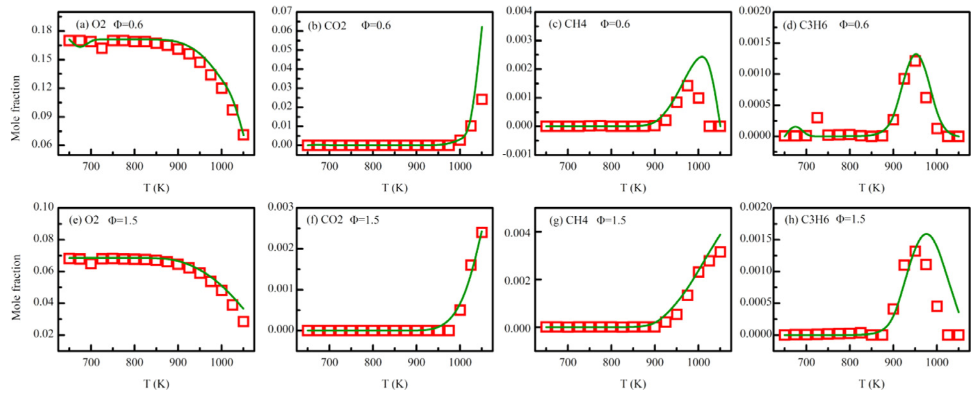


**Fig. 13.** The experimental and simulated speciation profiles for RP-3 oxidation in a plug flow reactor. Symbols: experimental data in [25]; lines: simulations.

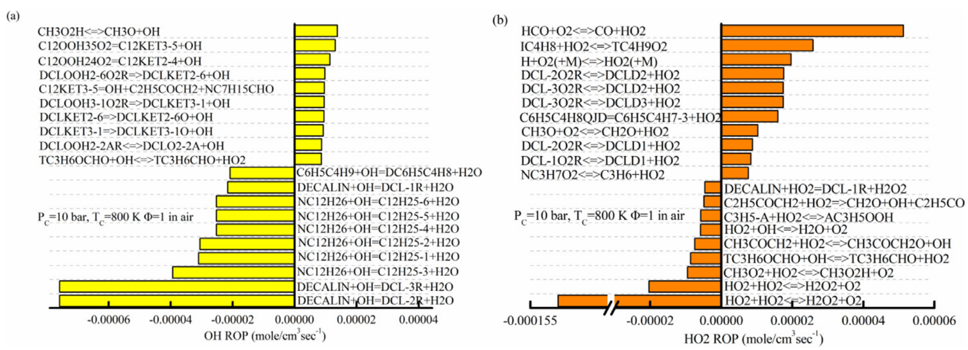


**Fig. 14.** Rate of production analysis of OH and $HO_2$ at 0.2 ms prior to the first stage ignition.

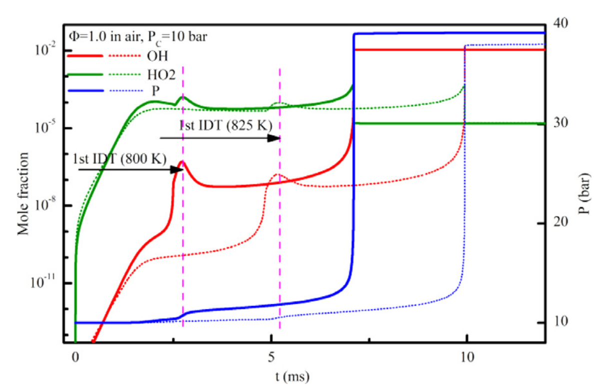


**Fig. 15.** Evolution histories of key radicals and pressure traces. Solid lines for 800 K, dot line for 825 K.

due to strongest reactivity, n-dodecane is the major OH consumer at the shown condition. On the other hand, the low temperature chain branching sequence of the isomerization of O2QOOH species and the subsequent decomposition of keto-hydroperoxide was the main OH formation channel. The significance of the cycloalkanes demonstrates that the exclusion of cycloalkanes in surrogate formulation is inappropriate. For $HO_2$, the recombination reaction, $HO_2$+$HO_2$=$H_2O_2$+$O_2$, is the major consumption channel. Through this routine, $H_2O_2$ accumulates until the temperature and pressure become sufficient for $H_2O_2$ to decompose into 2 OH radicals, causing the degenerate chain branching which dominate overall reactivity in NTC region. The concerted elimination reactions, e.g., DCL-2O2R=DCLD2+H2O, account for a great proportion of $HO_2$ generation, and these reactions are considered to be one of the reasons of NTC behavior happens. It must be pointed out that the rate of production of $HO_2$ is much higher than OH, and that's why the relative less reactive $HO_2$ plays an important role in NTC region.

The NTC behavior of the total IDT was fully studied by researchers, while the NTC behavior of the first stage IDT was not well confirmed and investigated because the first stage IDTs were very short in the possible temperature ranges and thus it is difficult to investigate it by experiments. However, kinetic analysis provides such an opportunity. Figure 15 displays the evolution histories of OH, HO2 and pressure for stoichiometric fuel/air mixture at 10 bar at temperatures of 800 K and 825 K, respectively. It is found that the mole fraction of OH and HO2 vary almost synchronously, although at different magnitude. Consistent with the ROP analysis, HO2 concentration is 5–6 orders of magnitudes higher than OH concentration. Two peaks are observed at both profiles at the two temperatures. Combining the pressure trace, it can be concluded that the two peaks correspond to the first stage and second stage ignition event. By comparing the OH, HO2 and pressure profiles at the two temperatures, it is found that the first-stage IDT at 825 K is obviously longer than the one at 800 K, demonstrating the NTC behavior of the first stage IDT. Other from the above-mentioned concerted elimination, the acceleration of other low temperature inhibition reactions with temperature, e.g., dissociation of RO2 and the decomposition of QOOH species into cyclic ether and alkene, are considered to be responsible for the NTC behavior of the first stage IDT.

## 6. Conclusions

In this study, a kinetic model was developed and validated for RP-3 based on a surrogate fuel proposed in this work. The surrogate was formulated for RP-3 using palette components within the typical molecular size of RP-3. Seven properties, including H/C, CN, MW, LHV, TSI, -CH3 and -CH2 was chosen as the optimized objective to determine the composition of the surrogate. Based on the proposed surrogate fuel, a kinetic model was developed to describe the combustion chemistry of RP-3 and validated against the database. The following results and conclusions are obtained:

(1) By the sequential use of a global and a local search method, a surrogate containing 27.44% n-dodecane, 28.81% iso-dodecane, 26.12% decalin and 17.63% n-butylbenzene by mole was developed for RP-3. The surrogate was used to compare with the target fuel, and the results showed that the two fuels agreed well with each other in terms of the target properties. Furthermore, the surrogate can satisfactorily mimic the key non-target properties of interest as well, e.g., density, surface tension and viscosity.
(2) The IDTs of the surrogate was measured in a heated RCM to further evaluate the performance the surrogate. The measurements were performed at pressures of 10, 15, 20 bar and equivalence ratios of 0.5, 1.0 and 2.0 over low-to-intermediate temperature ranges, exactly the same as the RCM experiments for RP-3. It is found that the surrogate accurately captured the variation trend of the IDTs with the operating condition parameters, and all the discrepancy of the magnitude basically falls within the error bar of 15%, further demonstrating the reasonability of the proposed surrogate.
(3) A detailed reaction mechanism was proposed by combining the previously developed sub-mechanisms of the components and core mechanism AramcoMech 2.0 to describe the combustion of the surrogate. By appropriate handling of the conflicts between sub-mechanisms, e.g., deletion of duplicated reactions and retaining the reliable datasets from the contradictory thermodynamic data and kinetic data sets, a kinetic model of 3065 species and 11,898 reactions was finally proposed. The mechanism was validated against the present data, as well as the experimental data in literature, and the results showed that predictions from the mechanism agree well with the experimental data.
(4) ROP analysis of OH and $HO_2$ were performed using the kinetic model for stoichiometric fuel/air mixture at 800 K to provide insight into the combustion in NTC region. The results show that concerted elimination reactions are related to the NTC phenomenon. Furthermore, evolution histories of the two species were simulated at 800 K and 825 K, respectively. The comparison implies that the first stage ignition has NTC behavior as well.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2021.111401.

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